VALUE ADDED PROBLEM 1:

Given a Binary Search Tree and a range [low, high]. Find all the numbers in the BST that lie in the given range. Note: Element greater than or equal to root go to the right side. Concept: Binary Search Tree

Example 1:

Input:

```
        17
      /    \
     4      18
   /   \
  2     9
l = 4, h = 24
```

Output:

4 9 17 18

Example 2:

Input:

```
        16
      /    \
     7      20
   /   \
  1     10
l = 13, h = 23
```

Output:

16 20

```
#include <stdio.h>
#include <stdlib.h>

// Node structure
struct node {
   int data;
   struct node* left;
   struct node* right;
};

// Create node
struct node* createNode(int value) {
   struct node* newNode = (struct node*)malloc(sizeof(struct node));
   newNode->data = value;
   newNode->left = NULL;
   newNode->right = NULL;
   return newNode;
}

// Insert into BST
struct node* insert(struct node* root, int value) {
   if (root == NULL)
      return createNode(value);

   if (value < root->data)
      root->left = insert(root->left, value);
```

```
    else
        root->right = insert(root->right, value);

    return root;
}

// Print nodes in range
void printInRange(struct node* root, int low, int high) {
    if (root == NULL)
        return;

    if (root->data > low)
        printInRange(root->left, low, high);

    if (root->data >= low && root->data <= high)
        printf("%d ", root->data);

    if (root->data < high)
        printInRange(root->right, low, high);
}

// Main
int main() {
printf("Bhavik Khobragade \nPRN = 25070521153 \nSEC = C2\n");
    // -------- Example 1 --------
    struct node* root1 = NULL;

    root1 = insert(root1, 17);
    root1 = insert(root1, 4);
    root1 = insert(root1, 18);
    root1 = insert(root1, 2);
    root1 = insert(root1, 9);

    printf("Example 1 Output: ");
    printInRange(root1, 4, 24);



    // -------- Example 2 --------
    struct node* root2 = NULL;

    root2 = insert(root2, 16);
    root2 = insert(root2, 7);
    root2 = insert(root2, 20);
    root2 = insert(root2, 1);
    root2 = insert(root2, 10);
```

```
    printf("\nExample 2 Output: ");
    printInRange(root2, 13, 23);

    return 0;
}
```

Output

```
Bhavik Khobragade
PRN = 25070521153
SEC = C2
Example 1 Output: 4 9 17 18
Example 2 Output: 16 20

=== Code Execution Successful ===
```